

|     |                                                                                                                                                                                                                                                                                       |                          |     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----|
| 10  | (a) product of density (of medium) and speed (of ultrasound) in the medium                                                                                                                                                                                                            | M1<br>A1                 | [2] |
| (b) | (i) $7.0 \times 10^6 = 1.7 \times 10^3 \times \text{speed}$<br>speed = $4.12 \times 10^3 \text{ m s}^{-1}$<br>wavelength = $(4.12 \times 10^3)/(9.0 \times 10^5) \text{ m}$<br>= 4.6 mm (2 s.f. minimum)                                                                              | C1<br><br>C1<br>A1       | [3] |
|     | (ii) for air/tissue boundary, $I_R/I \approx 1$<br>for air/tissue boundary, (almost) complete reflection/no transmission<br>for gel/tissue boundary, $I_R/I = 0.1^2/3.1^2$<br>= $1.04 \times 10^{-3}$ (accept 1 s.f.)<br>gel enables (almost) complete transmission (into the tissue) | M1<br>A1<br><br>M1<br>A1 | [4] |